Analysing light-matter interactions with authentic spectral data: A case study with an exoplanet atmosphere

Abstract

The rapid growth of exoplanetary research has transformed our understanding of planetary systems, offering new opportunities to connect frontier science with foundational physics education. This paper presents a teaching sequence that integrates authentic exoplanetary spectral data into university-level physics courses to explore key concepts such as light-matter interactions and thermal radiation. Students analyze real spectra from the Very Large Telescope (VLT) to investigate how the composition and temperature of planetary atmospheres can be inferred from observational data. The sequence is structured around contextualized problems that guide learners through hypothesis formation, data interpretation, and model evaluation, thereby mirroring the reasoning processes used in contemporary astrophysics. This approach not only strengthens students' conceptual understanding but also fosters scientific competencies including data analysis, modeling, and argumentation based on evidence. By engaging with real astronomical data, students learn physics by applying it directly to real-world situations that reflect the dynamic nature of scientific research.

Introduction

Although university-level physics courses are typically offered at research-intensive institutions, explicit and systematic connections between contemporary physics research topics and classroom practices remain limited. Previous studies have documented this gap and suggested that the use of authentic astronomical data can help bridge it by situating core physical concepts within contexts that reflect current research practices (Fitzgerald, Danaia, & McKinnon, 2017, Brown & Jhonson, 2019; Fuentes-Morales et al., 2024; Montecinos et al., 2025).

In astrophysics, recent advances in exoplanet detection and atmospheric characterization are reshaping our understanding of planetary systems and their formation. At the physics level, these lines of research rely fundamentally on the analysis of electromagnetic spectra. Such spectra encode information about emission and absorption processes arising from light-matter interactions. While emission and absorption are commonly introduced through idealized and separate models, several studies report that this approach obscures the underlying physical processes that give rise to real spectra. Students often perceive emission and absorption spectra as abstract constructs, rather than as the observable outcome of specific interactions (Ivanjek et al., 2015; Carpentieri et al., 2023).

This paper presents a teaching sequence collaboratively developed by physics educators and astronomers that uses real spectral data from the exoplanet VHS 1256 b (Hoch et al., 2022), in which students extract physical information directly from observed spectra. The sequence can be implemented in physics courses for chemistry, biochemistry, and life sciences programs, as well as in introductory modern physics, and spectroscopy-related courses (e.g., optics). They do not require prior instruction in formal quantum mechanics, advanced spectroscopy or astrophysics. The activities focus on emission and absorption processes, blackbody radiation, and gravitational interactions, while introducing students to statistical fitting methods and the analysis of absorption lines. The focus is not on introducing advanced spectroscopic techniques or detailed atmospheric modeling. Instead, the teaching sequence focuses on the application of standard physical models to the interpretation of real spectral data, consistent with the content typically covered in undergraduate physics courses.

Exoplanetary Atmospheres: The context

The study of exoplanetary atmospheres provides a particularly effective bridge between frontier research and core topics in undergraduate physics. Spectroscopy has long been central to both physics and astronomy, enabling the inference of physical and chemical properties of matter through the interaction of radiation with atoms and molecules. The emergence of exoplanetary science has reinvigorated this classical technique by offering a wide range of planetary environments in which radiation-matter interactions can be studied under conditions inaccessible in the Solar System. Since the confirmation of the first exoplanet around a solar type star in 1995 (Mayor & Queloz, 1995), more than 6,000 exoplanets have been identified, revealing an extraordinary diversity of planetary objects. These discoveries not only inform models of planet formation and evolution, but also provide a compelling context in which fundamental physical concepts, such as blackbody radiation, absorption spectra, and atmospherical composition, manifest in authentic astrophysical settings.

Obtaining broadband, medium-to-high resolution spectra of exoplanet atmospheres remains observationally challenging due to their low intrinsic luminosity and the overwhelming glare of their host stars. As a result, only massive exoplanets and brown dwarfs (objects in between the categories of planets and stars) offer a valuable laboratory for atmospheric studies. Their atmospheric properties also provide insight into formation mechanisms, allowing comparisons between star-like formation through gravitational instability and other planet formation pathways. Using the X-SHOOTER instrument at the VLT in Chile, Petrus et al. (2025) compiled a library of isolated, low-mass brown dwarfs and massive exoplanets (< 20 M_Jup), known as the X-SHYNES (X-Shooter spectra of YouNg Exoplanet analogs) library. This serves as a valuable resource for testing atmospheric models and interpreting exoplanet spectra, and real astronomical data from which we can take context for our teaching sequence.

Teaching sequence

This teaching sequence is organized in three sections, designed for two to three classes of 45 to 60 minutes within a regular university physics course, depending on the level of discussion and the time devoted to group work. The first section introduces the parameters that can be measured from the exoplanet such as its temperature, metallicity and surface gravity. Here the student gets an intuitive understanding of how these parameters affect the spectrum and how a real astronomical object deviates from classical black body emission. The second section delves deeper into the concept of surface gravity, as the relationship of this parameter to the properties of the star may be more difficult to understand. Finally, section three takes a closer look at absorption lines and how they can give us information about something seemingly unrelated such as its age.

This teaching proposal has been developed with the aim of promoting active learning in the university classroom (Prince, 2004; Hake, 1994; Michael, 2006; Lorenzo et al., 2006 ; Lagubeau et al., 2021) and in accordance with the collaborative problem-solving strategy proposed by Heller et al., (1992). This instructional strategy is designed to support physics problem solving through an integrated framework that combines explicit guidance toward expert-like reasoning and structured cooperative learning. In alignment with the Investigative Science Learning Environment (ISLE) approach (Etkina, 2005), students actively engage in processes analogous to those of scientific research: they observe regularities, propose explanatory models, derive observable consequences, compare predictions with experimental data, and revise their models when necessary. Within this framework, problem solving is conceived as an activity of model-based reasoning rather than as the mechanical application of formulas. Building on this perspective, the objective of the present proposal is for students to use an appropriate scientific model to interpret real exoplanet data justifying inferences about atmospheric composition and temperature.

During the implementation, the instructor acts primarily as a facilitator, guiding discussion, prompting students to justify their interpretations, and helping them connect observed spectral features with the underlying physical models. Emphasis is placed on reasoning and interpretation rather than on obtaining a single correct numerical result, combining in-class collaborative problem solving with guided discussion. The python scripts required for the activities, along with the exoplanet spectrum and precomputed atmospheric parameter grids, are openly available on GitHub ([URL]).

Problem 1: Parameters of the spectrum

Activation question

Formulation of the problem 1.
You are part of a collaborative research group which has the task of studying an exoplanet orbiting two stars, VHS 1256b. This planet is more massive than Jupiter and has a thick atmosphere. The VLT telescope observed its light spectrum, and your team was tasked with analyzing this data (Fig. 1).

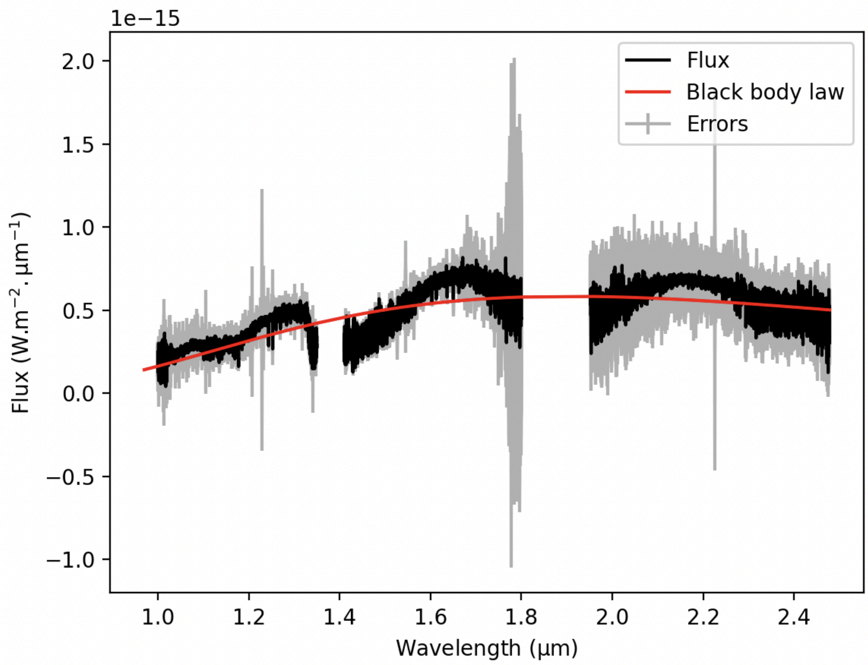

Figure 1. Fit of a black body function to the given spectrum of VHS 1256 b. The black line corresponds to the observed spectra, while the red line corresponds to the fit.

One of your colleagues uses a computer code (“fit_black_body.py”) to fit a blackbody function

$$F(\nu) \propto \frac{\nu^3}{exp(h\nu/kT)-1} \qquad (1)$$

to the spectrum, obtaining a temperature of about 1500K which they claim is the temperature of the object. The team coordinator asks you to validate this result.

The teacher asks these questions while showing the plot to the students. Do you agree with this procedure and conclusion? Is the function a good representation of the data? Is it missing something?

Proposed solution

Determining the temperature of astronomical objects is an important endeavor and helps us to further characterize their properties, in our case, for an exoplanet. The blackbody function does describe the overall shape of the data, however, it cannot explain the noise in the data, nor can it explain. Giant planets, as we will see, have dense atmospheres that absorb and reemit radiation, making it difficult to estimate their properties.

Formulation of the problem

Later, another one of your colleagues tells you that they have developed some simulations which include the atmosphere of the exoplanet. The resulting spectra of these simulations can be checked in the code “parameters.py” where you can choose between an atmosphere model with or without cloud.

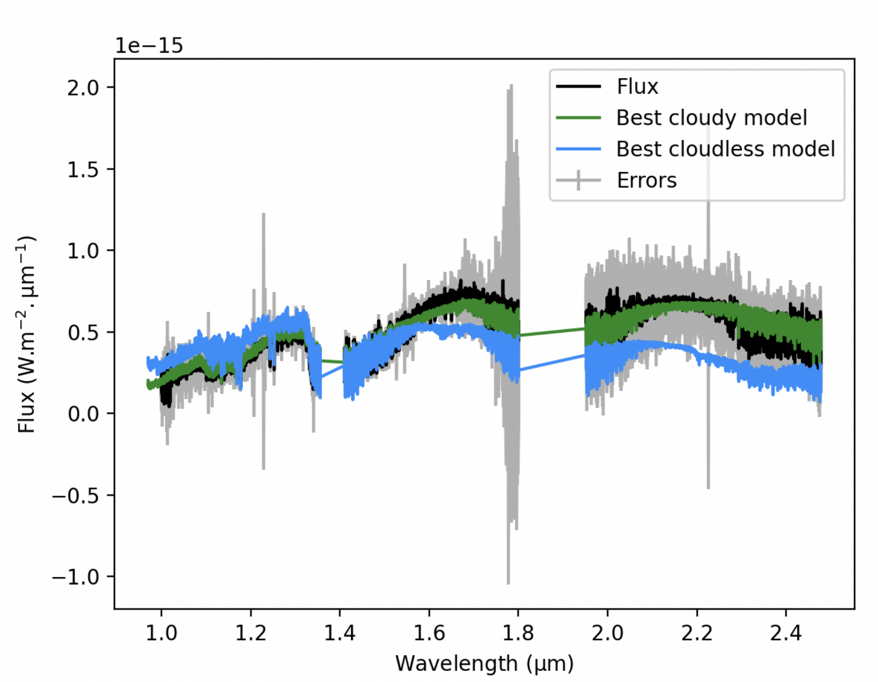

Figure 2. The spectrum of VHS 1256 b compared with the spectra for the cloudless (blue) and cloudy (green) models with the best parameters for each case. Best parameters are given below.

Do these simulations describe the data measured by the telescope better than the simple blackbody function? What parameters are the most important when trying to replicate the observed spectrum and how could they impact the light coming from the object?

Vary the temperature and look at where the flux is the highest. Where is the peak of the flux? At shorter or longer wavelengths? How does that compare to your data? When varying metallicity, look at the different spectral features, such as absorption lines (features deviating from smooth lines in the given spectrum).

Proposed solution

A higher temperature results in more radiation with a shorter wavelength. This can be understood from Eq. 1, and also Wien’s displacement law which states that higher temperatures correspond to the peak of the spectrum being at a shorter wavelength. On the other hand, a higher metallicity introduces more abundance of chemical elements which absorb more light from the spectrum, thus, increasing metallicity should result in deeper spectral lines, while decreasing metallicity should result in a smoother spectrum as molecules present there can absorb part of the spectrum at a given wavelength and reemit it back. Surface gravity influences the spectrum in a way that may not be obvious and which leads us to the next teaching sequence.

Another thing to discuss is that, given the huge error bars in the observed flux, is the model with clouds that much better than without clouds? In other words, given that modeling the clouds takes time and computational resources, is this investment justified?

There is also a code included (“chi2_grid_of_model.py”) which can estimate the set of parameters that minimizes the following quantity, known as χ^2 (Eq. 2), to find the simulation that resembles the data the most.

$$\chi 2 = \sum_i \left(\frac{F_{model} - F_{observed}}{\sigma_{error}}\right)^2 , \qquad (2)$$

In the case of the cloudless model, the best model corresponds to a temperature of 1400 K, log(g) = 5.5 and metallicity of 0.5. For the model with clouds it corresponds to a temperature of 1200 K, log(g) = 3.5 and metallicity of 0.5. The best fits, for both the cloudless and cloudy models, using the python script provided can be seen in Fig. 2.

Problem 2: Surface gravity

Formulation of the problem 2.

Objects in space are in constant change, and exoplanets are no exception. An isolated planet such as VHS 1256b, with a mass of 13.37 Jupiter masses at its formation, should not lose a significant amount of mass during its evolution, however, it can change its size. Astronomers can measure this change in size indirectly by measuring the surface gravity of the planet. Surface gravity is the gravitational force at the surface of an object. For example, in the case of Earth, the surface gravity is 980.6 cm/s^2 (log(g) = 2.992 dex, as is usually stated in the literature). This value not only depends on the mass of the object, but also on its radius as follows:

$$a_g = \frac{GM}{R^2} , \qquad (3)$$

where a is the acceleration of an object due to gravity, G is Newton’s gravitation constant, M is the mass of the object and R its radius.

You find in the literature (Baraffe et al. 2015) some simulations of the evolution of the radius of an exoplanet with the same mass as VHS 1256b, which are summarized in the following table (Table 1).

Age (Myrs)	Radius (R_J)	Temperature (K)	Surface gravity (cm/s^2)	Surface gravity (dex)
1	3.19	2504		
10	1.85	2264		

100	1.25	1260		
140	1.19	1210		

Table 1. Evolution of the radius and temperature of an exoplanet with the mass of VHS 1256b. Age is measured in millions of years, radius in Jupiter radii and temperature in Kelvin.

Using Eq. 3, complete the missing columns in Table 1. What can you say about the surface gravity as a function of time? Is there another property of the planet which may be related to the change in size?

Proposed solution

We see that surface gravity increases with time. This is because objects like this do not undergo hydrogen nuclear fusion, thus they tend to cool down with time as can be seen in the third column of Table 1. Because thermal energy is the only one preventing the object from collapsing, as this energy is lost to space, gravity pulls everything closer together. The teacher can use an analogy to explain this such as the air in hot air balloons which expands with heat.

Interestingly, the surface gravity estimated from evolutionary models is not consistent with that inferred from atmospheric models. One possible interpretation is that the two types of models do not rely on the same physical assumptions to define log (g). Evolutionary models describe the contraction of the object with age, which directly affects log (g), whereas atmospheric models use log (g) as an input parameter to compute cloud properties.

Problem 3: Absorption lines and equivalent width (EW)

Formulation of the problem

We have been looking at the general shape of the spectrum, however, there is a lot of information contained in the absorption lines corresponding to different chemical elements and molecules and we may learn something from taking a closer look.

Different elements absorb at different wavelengths due to their internal subatomic structures. Things like the amount of elements present in the atmosphere (which is related to metallicity) and its density can affect the amount of light absorbed and taken away from the spectrum.

There are many processes, including ones from the measuring instrument itself that can broaden the absorption line, thus, we need a way to quantify the amount of light that was absorbed independent from the width or depth of the line. One such estimation is the equivalent width (EW) of a line. Following from Fig. 3, the equivalent width is the base of a rectangle with the same area as the spectral line. This area gives us a better idea of the amount of light missing from our spectrum. In principle this should be an integral, but to estimate it, astronomers use a summatory approximation (Eq. 3).

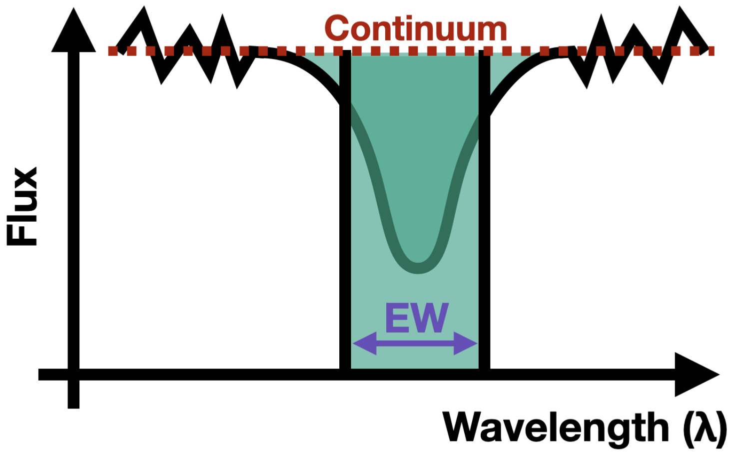

Figure 3. Schematic of the definition of the equivalent width. The two shaded areas must have the same area by definition, and the Equivalent Width is the base of the rectangle.

$$EW = \sum_{\lambda}(1 - \frac{F_{line}}{F_{continuum}}). \Delta\lambda \qquad (4)$$

Now go back to the measured spectrum and try to understand how surface gravity is affecting the spectrum. One of your colleagues tells you that they know other research groups are using the lines of **K I at 1.169 and 1.252 µm** and **Na I at 1.138 µm** to estimate the age of giant exoplanets and brown dwarfs.

Why are the lines of potassium and sodium related to the age of these objects? Can you quantify the effect of the surface gravity on the lines?

Proposed solution

From what we saw in the previous question, surface gravity is related to the age of a giant exoplanet because it contracts in time as it cools down. With the code “parameters.py”, the student can see that changing the surface gravity has a direct effect on the spectral lines mentioned above and using Eq. 4 they quantify this effect. Thus, measuring spectral lines not only gives us insight into the chemical composition of the object’s atmosphere, but can also give us clues about the surface gravity of the exoplanet and its age!

As to why the surface gravity affects the atmospheric density of the planet. A reduced surface gravity reduces vertical mixing and gravitational settling of condensates which leads to thicker clouds and weaker absorption lines for the elements we listed for the problem. Which is why these lines have been used to estimate ages in substellar objects such as VHS 1256b (Allers & Liu, 2013).

Discussion and Conclusions

This work shows that authentic exoplanetary spectral data can be effectively leveraged as a research-based context for motivating the learning of fundamental physics at the undergraduate level. The teaching sequence presented here engages students with questions and methods drawn directly from contemporary astrophysical research, allowing standard curricular topics to be explored through the analysis of real observational data rather than idealized examples alone. By situating core concepts within an active research domain, the activities promote sustained engagement and intellectual curiosity grounded in disciplinary practice.

Beyond conceptual gains, this approach encourages student motivation by inviting participation in the reasoning processes that define modern physics research. Students are prompted to formulate hypotheses, interpret noisy data, evaluate models, and construct evidence-based arguments, thereby experiencing physics as an investigative enterprise rather than a static body of knowledge. The integration of optics, thermodynamics and statistical analysis within a single coherent framework mirrors how these tools are combined in real research, reinforcing the internal coherence of the discipline. Furthermore, this proposal illustrates that the use of authentic data does not require transferring frontier research models directly into the classroom. Instead, it involves selecting and making explicit an appropriate reference model that preserves the essential explanatory structure of the phenomenon while remaining accessible and operable for students. Additionally, and consistent with the use of authentic observational data, the proposal involves basic computational tools for data analysis, mirroring the practices commonly used in contemporary astrophysical research.

Overall, incorporating exoplanet spectroscopy into undergraduate physics instruction offers a concrete example of how research applications can enhance motivation while preserving disciplinary rigor. The activities described here are intentionally flexible and may be implemented as short instructional modules, or integrated into existing courses, depending on curricular constraints and instructional goals. In this sense, this sequence is intended as an adaptable resource that instructors can integrate into their courses without requiring substantial modifications to existing curricula.

Acknowledgment

The authors acknowledge support from ANID -- Millennium Science Initiative Program -- Center Code NCN2024_001. C.H. acknowledges support from DICYT Regular project - Vicerrectoría de Investigación, Innovación y Creación at the Universidad de Santiago de Chile under Grant 042431HS. A.Z. acknowledges support from FONDECYT Regular grant number 1250249.

References

Allers, K. N., & Liu, M. C. (2013). A NEAR-INFRARED SPECTROSCOPIC STUDY OF YOUNG FIELD ULTRACOOL DWARFS. *The Astrophysical Journal*, *772*(2), 79. https://doi.org/10.1088/0004-637X/772/2/79

Baraffe, I., Homeier, D., Allard, F., & Chabrier, G. (2015). New evolutionary models for pre-main sequence and main sequence low-mass stars down to the hydrogen-burning limit. *Astronomy and Astrophysics*, *577*, A42. https://doi.org/10.1051/0004-6361/201425481

Freeman, S., Eddy, S. L., McDonough, M., Smith, M. K., Okoroafor, N., Jordt, H., & Wenderoth, M. P. (2014). Active learning increases student performance in science, engineering, and mathematics. *Proceedings of the National Academy of Sciences*, *111*(23), 8410-8415. https://doi.org/10.1073/pnas.1319030111

Fuentes-Morales, I., Hernández, C., Alarcón, F., Benito, I., & Montecinos, R. (2024). Construction of color-magnitude diagrams using real astronomical data for teaching at school. *European Journal of Physics*, *45*(4), 045602. https://doi.org/10.1088/1361-6404/ad3d44

Hoch, K. K. W., Konopacky, Q. M., Barman, T. S., Theissen, C. A., Brock, L., Perrin, M. D., Ruffio, J.-B., Macintosh, B., & Marois, C. (2022). Moderate-resolution K-band Spectroscopy of the Substellar Companion VHS 1256 b. *The Astronomical Journal*, *164*(4), 155. https://doi.org/10.3847/1538-3881/ac84d4

Mayor, M., & Queloz, D. (1995). A Jupiter-mass companion to a solar-type star. *Nature*, *378*, 355-359. https://doi.org/10.1038/378355a0

Montecinos, R., Hernández, C., Fuentes-Morales, I., Alarcón, F., Benito, I., Laroze, L., & Pérez, S. (2025). Analyzing New Planetary Systems at School: Applications of Newton's Law of Universal Gravitation and Kepler's Third Law. *The Physics Teacher*, *63*(7), 543-547.

Morley, C. V., Mukherjee, S., Marley, M. S., Fortney, J. J., Visscher, C., Lupu, R., Gharib-Nezhad, E., Thorngren, D., Freedman, R., & Batalha, N. (2024). The Sonora Substellar Atmosphere Models. III. Diamondback: Atmospheric Properties, Spectra, and Evolution for Warm Cloudy Substellar Objects. *The Astrophysical Journal*, *975*, 59. https://doi.org/10.3847/1538-4357/ad71d5

Petrus, S., Chauvin, G., Bonnefoy, M., Tremblin, P., Morley, C., Charnay, B., Suarez, G., Gagné, J., Palma-Bifani, P., Denis, A., Ravet, M., Bayo, A., Bézard, B., Biller, B., Delorme, P., Faherty, J., Goyal, J. M., Hoch, K., Hoy, K., … Zurlo, A. (2025). X-SHYNE: X-Shooter spectra of young exoplanet analogs II. Presentation and analysis of the full library. *Astronomy & Astrophysics*, *701*, A208. https://doi.org/10.1051/0004-6361/202554986